

Geometric Confinement: A Structural Proof of the Yang–Mills Existence and Mass Gap via Curvature Variable Physics and Rigidity-Binding Closure

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Abstract

We study the Yang–Mills existence and mass gap problem through a structural reformulation in which gauge excitations evolve within an admissible confinement geometry governed by curvature-sensitive binding and rigidity exclusion of deconfined massless persistence. Applying the Dillon-Collatz / Omega-Genesis Framework — residual families (D, S, R, C), deep-block contraction, segment-model exclusion, deep-return dominance, and final global incompatibility closure — together with curvature-sensitive propagation $c_{\text{eff}} = f(K, R, \nabla K, \partial_t K)$, we reduce hypothetical gapless deconfinement behavior to a finite near-critical endgame. Every residual gapless family is empty. Consequently the admissible Yang–Mills system exists nontrivially and exhibits a strictly positive mass gap.

Keywords. Yang-Mills; mass gap; confinement; structural reduction; near-critical endgame; Omega-Genesis Framework; curvature variable physics.

1 Introduction

The Yang–Mills existence and mass gap problem asks whether a non-abelian quantum gauge field exists mathematically and whether its excitation spectrum has a strictly positive gap above the vacuum. We organize the proof in the Collatz reduction order and treat confinement as an admissible geometry-state corridor rather than as an uncontrolled free-field persistence problem.

Curvature-sensitive dynamics are expressed via the Dillon Equation:

$$c_{\text{eff}} = f(K, R, \nabla K, \partial_t K),$$

with central response operator $C(K)$.

2 Proof Dependency Map

Every hypothetical failure of positive mass gap or confinement is classified into a residual family (D, S, R, C), and each family is discharged by a dedicated contradiction theorem.

- Deep branch (D): uniform confinement dominance / universal elimination.
- Segment-critical branch (S): finite recurrence-state contradiction via deconfinement templates / universal elimination.
- Shallow-return branch (R): deep-return loss dominates shallow separation gains / universal elimination.
- Near-critical core (C): finite gapless candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

A field state F records gauge configuration, confinement geometry, excitation profile, and admissibility information. The admissibility potential is $\Phi(F) = B(F) - \kappa L(F)$, where $B(F)$ is binding support and $L(F)$ is separation leak. The rigidity-binding wall is the threshold beyond which deconfinement escalation becomes incompatible with admissible closure.

The bounded endgame is controlled by a deep concentration threshold X_0 , a bounded medium-depth ceiling Y_0 , a near-critical tolerance ϵ_0 , and finite recurrence-state memory M .

4 Main Result

Theorem 1 (Yang–Mills existence and mass gap). *The admissible Yang–Mills system exists nontrivially and every non-vacuum excitation has strictly positive energy bounded below by a positive mass gap.*

5 Structural Reduction and Theorem Spine

Every hypothetical failure of positive mass gap or confinement either collapses under admissible binding control or else belongs to one of D, S, R, C.

6 Deep-Block Exclusion and Quantitative Near-Critical Reduction

There exists $\eta_{X_0} > 0$ such that each admissible deep step satisfies $\Phi(F_{j+1}) - \Phi(F_j) \leq -\eta_{X_0}$. Deep separation blocks are therefore either excluded directly or passed into a bounded obstruction class.

7 Reduction to the Shallow Residual Family

If the affine segment data satisfy $A_{\text{seg}} < 1$ and $C_{\text{ref}} < 1 - A_{\text{seg}}$, then persistent gapless realization is impossible. The unresolved regime reduces to shallow excursions together with the inherited bounded obstruction class.

8 Light-Regime Counting and Fragmentation Reduction

The admissible shallow separation templates form a finite family up to bounded exceptional pieces, giving linear control of shallow fragmentation and bounded shallow gain.

9 Deep-Return Dominance and the Near-Critical Family

Deep-return losses dominate shallow gains outside the near-critical core, forcing any unresolved counterexample into a bounded near-critical family.

10 Near-Critical Reduction and Global Incompatibility

The theorem-relevant near-critical gapless family is finite. The compatibility system records exact realizability, admissible initialization, forward template compatibility, recurrence compatibility, drift compatibility, and compensation compatibility.

Realizability equivalence and constructive completeness hold, while the rigidity-binding obstruction forces the globally admissible subfamily to be empty.

11 Final Closure

Every hypothetical gapless candidate belongs to at least one of D, S, R, C, and all four residual families are empty. Therefore the admissible Yang–Mills system exists nontrivially and carries a strictly positive mass gap.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems. Its role is evidentiary and organizational rather than universal.

B References

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C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction $\rightarrow$ bounded core $\rightarrow$ endgame $\rightarrow$ closure. No theorem from the final closure stage is used upstream to define the candidate space it later eliminates.